Indian Institute of information Technology Sonepat

(Operating System)(IT)(ITC 403)

Mid Term-I , Date: Feb 27, 2025

12312021

Timing: 9:30 to 10:30 AM | IV Semester | Max mark: 15

All questions are Compulsory

1. (a) Explain system calls in operating systems and categorize them with examples.. (2.5)

   (b) Explanation of Architecture That Defines Working Functions of an Operating System: (2.5)

2. Explain the concepts of process management and memory management in an operating system. Discuss the different states of a process and the process lifecycle.' (5)

3. Explain the concept of a kernel in an operating system. Compare and contrast the different types of kernels. (5)

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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
SONEPAT

B.Tech 4th Sem ,IT Mid-Semester Examination, Apr. ,24, 2025.

Subject: Operating System **Subject Code: ITC 403**

Roll No 12312021

Branch: IT **Max.Marks.-15 (3+4+8)** **Time- 1 Hour**

Note: Related Figures/Diagram/ DFD is mandatory in all the respective questions.

1. Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. The contents of the frames are as follows initially and after each reference (from left to right):

initially	*after*												
-	1*	2*	3*	4*	5*	4	1	6*	4	5	1*	3*	2*
-	1	1	1	1	1	1	1	6	6	6	6	6	2
-	-	2	2	4	4	4	4	4	4	4	1	1	1
-	-	-	3	3	5	5	5	5	5	5	5	3	3

The * marked references cause page replacements.
Which one page replacement policy/policies could be the in use?

2. Apply LRU and Optimal page replacement policies for the following reference string
and calculate how many page-faults would occur for three-page frame structure;
1, 2, 3, 4, 5, 5, 3, 4, 9, 6, 2, 7, 8, 7, 3, 5, 9, 7, 6, 4, 5, 4, 2.

3. Define the working structure of following algorithms:
   a) Banker's algorithm
   b) Dining Philosopher Problem
   c) Reader Writer Algorithm
   d) Paging and Segmentation

Roll No.	1	2	3	1	2	0	2	9

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

Department of Information Technology (IT)

Microprocessor and Interfacing (UCS015)

Mid Sem Exam-I (February 28, 2025)

Branch: IT   Semester-IV   Max marks: 15

(All questions are compulsory)

1. Explain the architecture of 8085 microprocessor with its key features in details. (5)

2. (a) How the lower-order address lines and the data lines ($AD_7 - AD_0$) can be de-multiplexed? Explain it with the help of a diagram. (2)

   (b) Define the following terms: (3)

   (i) Instruction and T-state

   (ii) Program Status Word

   (iii) Program Counter (PC)

3. Consider an 8085 microprocessor system, the following program starts at locaton 0100H. Assume that the content at memory location 0207H is 60H and all flags are reset initially. (5)

```
LXI SP, 00FFH
LXI H, 0207H
MVI A, 20H
SUB M
```

   (i) What is the content of accumulator when the program counter reaches 0109H ?

   (ii) How many bytes are required to store the aforementioned lines of the code?

   (iii) Discuss types of addressing modes, number of machine cycles, number of T-states, and the status of each flag after executing the each line of code.

(End of the question paper)

Roll No. | 1 | 2 | 3 | 1 | 2 | 0 | 2 | 1 |

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Email: sonepatiiit@gmail.com, website: https://iiitsonepat.ac.in

Mid Sem-I | **Subject: Compiler Design (UIT406)** | **Roll no** 12312021

Branch: IT (4th Sem) | **M.M.-15 marks** | **Time- 60 mint.**

Note: All questions are compulsory.

Q1 Explain the conversion process of HLL (High Level Language) to Low Level Language (or Machine Code). Also discuss different phases of a compiler by taking an example of instruction as **(float a= b+c*60;).** — 5

Q2 Consider the context-free grammar

- $E \rightarrow E + E$
- $E \rightarrow (E * E)$
- $E \rightarrow \text{id}$

where E is the starting symbol, the set of terminals is $\{id, (, +,), *\}$, and the set of non-terminals is $\{E\}$.

For the terminal string $id + id + id + id$, how many parse trees are possible? — 3

Q3 Check whether the grammar is ambiguous or not with proper explanation — 3

S → AB / C

A → aAb / ab

B → cBd / cd

C → aCd / aDd

D → bDc / bc

Q-4 Discuss the process of removing of left recursion and left factoring in the grammar by taking an example. Consider the following grammar and eliminate left recursion and left factoring. — 4

S → A

A → Ad / Ae / aB / ac

B → bBc / f

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Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

B.Tech. Midterm (2) Examination, April'2025.

Subject: Computer Networks | **Subject Code: ITC404**

Roll No. 1 2 3 1 2 0 2 1

Branch: IT | **Max.Marks.-15** | **Time- 1 Hours**

Note: All questions are compulsory.

Q-1	a) Distinguish ARP and RARP protocols and their services. b) State the difference between classless and classful addressing. c) Differentiate router and bridge.	(1.5+1.5+2)=5 Marks
Q-2	a) Explain the difference between network layer delivery and transport layer delivery. b) How to improve QoS in the transport layer. c) Explain Token Bucket Protocol.	(1.5+1.5+2)=5 Marks
Q-3	a) Write a Brief Notes on following (*any 2*) i) RIP ii) Datagram iii) Segment iv) UDP b) What is the purpose of the Domain Name System?	((2+2)+1)=5 Marks

60

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Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

B.Tech. Midterm (10 Examination, Feb'2025.

Subject: Computer Networks

Subject Code: ITC404

Roll No. 12312021

Branch: IT

Max.Marks- 15

Time- 1 Hours

Note: All questions are compulsory, and your answer should include a rough sketch, if necessary.

Q-1	a) Define data communications. Why are standards needed? b) Explain different types of topology. c) *A line has a signal-to-noise ratio of 2000 and a bandwidth of 5000 KHz. What is the maximum data rate supported by this line?*	(1.5+1.5+2)=5 Marks
Q-2	a) Describe the OSI layer's function using a suitable sketch diagram. b) Explain PPP frame format. What is the MAC address and how is it related to NIC? c) Point out network layer responsibilities in OSI model.	(2.5+1.5+1)=5 Marks
Q-3	a) What is a sliding window (ARQ)? Describe with the help of the diagram the *Go-Back-N* continuous ARQ error control scheme. b) What is an error? Explain different type of error. c) Sketch the Manchester and differential Manchester for bit stream:0001110101.	(2.5+1.5+1)=5 Marks

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Email: sonepatiiit@gmail.com, website: https://iiitsonepat.ac.in

Mid Sem-II | **Subject: Compiler Design (UIT-406)** | **Roll no** 12312021

Branch: IT (4th Sem) | **M.M.-15 marks** | **Time- 60 mint.**

Note: All questions are compulsory.

Q-1 Consider the following context-free grammar where the set of terminals is {a,b,c,d,f}.

$$S \rightarrow d\,a\,T \mid R\,f$$
$$T \rightarrow a\,S \mid b\,a\,T \mid \epsilon$$
$$R \rightarrow c\,a\,T\,R \mid \epsilon$$

The following is a partially-filled LL (1) parsing table.

	a	b	c	d	f	\$
S			(1)	$S \rightarrow d\,a\,T$	(2)	
T	$T \rightarrow a\,S$	$T \rightarrow b\,a\,T$	(3)		$T \rightarrow \epsilon$	(4)
R			$R \rightarrow c\,a\,T\,R$		$R \rightarrow \epsilon$	

Fill the appropriate productions LL (1) parsing table in the location (1), (2), (3) and (4).

Q-2 Construct LL(1) parsing table of the following grammar:

$S \rightarrow aBDh$ $B \rightarrow cC$ $C \rightarrow bC/\varepsilon$ $D \rightarrow EF$ $E \rightarrow g/\varepsilon$ $F \rightarrow f/\varepsilon$

Also generate parse tree for the string "acbh".

Q-3 Construct LR(0) parsing table of the following grammar:

$S \rightarrow (\,L\,) \mid a$

$L \rightarrow L\,,\,S \mid S$

Q-4 Construct LR(0) parsing table of the following grammar and generate parse tree for the string "id+id+id".

$E \rightarrow T+E/T$

$T \rightarrow id$

Indian Institute of information Technology Sonepat
(Software Engineering)(IT)(UCS011)
Mid Term-II , Date: April 23, 2025

12312021

Timing: 9:30 to 10:30 AM | IV Semester | Max mark: 15

All questions are Compulsory

1. (a) A product development team is building a mission-critical software where requirements are not clearly defined at the outset but are expected to evolve through constant user feedback. However, the software must also undergo extensive documentation and review cycles due to regulatory compliance. Identify and justify the selection of the most suitable software development model for this scenario. Critically evaluate its strengths and limitations in the given context. (3)

   (b) Can you apply the Waterfall Model to the above scenario, even if the development team has extensive experience with it? Justify your answer with two strong technical reasons. (3)

2. You are a consultant advising a multinational product company. They are unsure which model to follow and give you this table of their product environment:

Factor	Observation
Requirement Stability	Partially stable; frequent mid-project shifts
User Involvement	High; user stories are evolving weekly
Risk Level	Moderate
Team Structure	Distributed across 3 time zones
Release Frequency	Every 2 weeks
Documentation Requirement	Medium, not regulatory

   (a) How would **pure Waterfall** or **Spiral** models create bottlenecks in such an environment? . (

Indian Institute of information Technology Sonepat

(Software Engineering)(IT)(UCS011)

Mid Term-I, Date: Feb 25, 2025

1231202l

Timing: 12:00 to 1:00 PM | IV Semester | Max mark: 15

All questions are Compulsory

1. (a) Identify and explain three key factors that contribute to the Software Crisis. Discuss how each factor impacts software development processes and outcomes. (2.5)

   (b) Suggest and explain two solutions that could help mitigate the Software Crisis. How do these solutions address the identified factors, and what benefits do they provide in modern software development. (2.5)

2. Discuss the disadvantages of using the *goto* statement in terms of readability, maintainability, and debugging. Illustrate these drawbacks with a program. How can these issues be overcome in modern programming languages? (5)

3. Explain the realization relationship in UML and its significance in object-oriented design. Write a program that demonstrates the realization relationship between an interface and its implementing classes. Also, include the UML diagram that illustrates this relationship. (5)

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(Total pages: 1)

Roll No.	1	2	3	1	2	0	2	1

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Department of Information Technology (IT)

Microprocessor & Interfacing (UCS015)

Mid Semester Examination-II (April 25, 2024)

Branch: IT　　Semester-IV　　Total marks: 15

(Note: All questions are compulsory)

1. (a) Explain the following instructions with example in detail. (2)

   i DAD r_p

   ii CALL add_{16}^{r}

   (b) In an 8085 microprocessor, the contents of the accumulator and the carry flag are A7 (in hex) and 0, respectively. What will be the contents of the accumulator (in hex) and the carry flag, respectively, if the instruction RLC is executed three times. (2)

2. (a) In what way TRAP is different from the other four hardware interrupts? (2)

   (b) Draw the RIM instruction format and discuss it. (2)

3. (a) Write a BSR control word to set bits PC_7 and PC_0 and to reset them after 1 second delay if the address of control word register (CWR) is 93H. (2)

   (b) Write down the main features of 8259. (2)

4. (a) An Intel 8085 microprocessor is executing the program given below. (3)

```
        MVI A, 20H
        MVI B, 10H
BACK:   NOP
        ADD B
        RLC
        JNC BACK
        HLT
```

What is the number of times that the instruction NOP will be executed and what will be the content of accumulator register (A) after executing HLT instruction ?

End of question paper !